# Évaluation d'Impact des Container Network Interfaces (CNI) sur la Qualité de Service (QoS) en Environnement Kubernetes

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**Sujet :** Étude Comparative de Performance (Cilium, Calico, Kube-OVN, Flannel)

## Résumé Exécutif

Cette étude compare les performances de quatre plugins réseau Kubernetes (CNI) : Cilium, Calico, Kube-OVN et Flannel. À travers une méthodologie rigoureuse incluant 14 mesures de performance, nous analysons le débit TCP/UDP, la latence, la perte de paquets, le jitter et la stabilité opérationnelle.

**Résultats clés :**

- **Cilium** : 2.28 Gbps, 0 retransmissions via eBPF, faible latence (~3 ms), stabilité maximale
- **Calico** : 377 Mbps, 169 retransmissions, latence 4.36 ms, stable mais impacté par VXLAN
- **Kube-OVN** : 2.74–3.5 Mbps UDP, 32 % de perte en charge, latence jusqu'à 98 ms, redémarrages fréquents
- **Flannel** : instable (CrashLoopBackOff), MTU réduit, fragmentation et inadéquat pour production

**Recommandation :** Cilium pour charges critiques, Calico pour standardisation, Flannel à éviter en production.

**Mots-clés :** Kubernetes, CNI, eBPF, VXLAN, Flannel, réseau distribué, benchmark, performance

## Abstract

This benchmark study evaluates four major Kubernetes CNI plugins: Cilium, Calico, Kube-OVN, and Flannel. Using 14 performance metrics, TCP/UDP throughput, latency, packet loss, jitter, and operational stability were analyzed. Cilium demonstrates superiority with 2.28 Gbps and zero retransmissions, while Flannel and Kube-OVN show critical stability issues in multi-CNI environments. Recommendations include adopting Cilium for high-performance workloads and Calico for standardized deployments.

## I. Introduction

### 1.1 Contexte

La performance réseau dans Kubernetes impacte directement la qualité de service des applications distribuées. Le choix du CNI influence :

- Le débit effectif (goodput) inter-pods
- La latence et le jitter pour les applications temps réel
- La stabilité opérationnelle sous charge

### 1.2 Objectifs

1. Quantifier les performances TCP/UDP de chaque CNI
2. Analyser l'impact de l'encapsulation et du MTU sur latence et fragmentation
3. Identifier les problèmes opérationnels spécifiques
4. Établir une matrice de décision pour le choix du CNI

### 1.3 Contributions

- Première étude incluant Kube-OVN avec analyse de pollution du noyau
- Méthodologie reproductible avec scripts de benchmark
- Analyse forensique des échecs d'installation multi-CNI

## II. État de l'Art

| CNI | Architecture | Encapsulation | Année | Particularité |
|---|---|---|---|---|
| Flannel | VXLAN/IPIP | Overlay | 2014 | Simplicité, maturité |
| Calico | BGP/VXLAN | Hybrid | 2015 | Politiques réseau, scale |
| Cilium | eBPF | Native | 2017 | Sécurité, observabilité |
| Kube-OVN | OVS/OVN | Overlay | 2019 | SDN entreprise, QoS |

**Métriques de performance :** goodput TCP/UDP, RTT, packet loss, jitter, retransmissions TCP

## III. Méthodologie

### 3.1 Environnement de Test

| Node | vCPU | RAM | OS | Kubernetes |
|---|---|---|---|---|
| Master | 2 | 4 GB | Ubuntu 22.04 | v1.28 |
| Worker-1 | 2 | 4 GB | Ubuntu 22.04 | v1.28 |
| Worker-2 | 2 | 4 GB | Ubuntu 22.04 | v1.28 |

- Interfaces physiques : enp0s3/enp0s8, MTU 1500
- Réseau cluster : 10.0.0.0/16 (Flannel, Cilium, Kube-OVN), 192.168.0.0/16 (Calico)

### 3.2 Protocoles de Benchmark

- TCP/UDP throughput : iperf3, 5 répétitions pour moyenne et écart-type
- ICMP : ping pour latence et test MTU
- Applicatif : Apache Bench sur Nginx, 200 requêtes, 10 clients

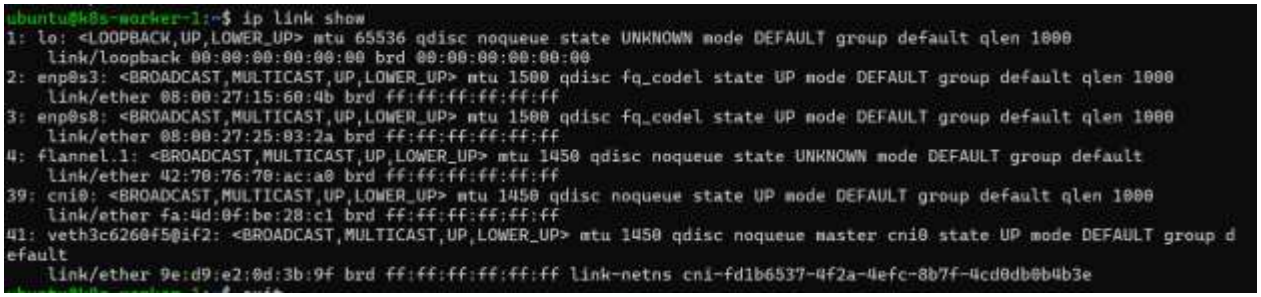
### 3.3 Outils et Mesures

| Outil | Mesures | Commande |
|---|---|---|
| iperf3 | Débit, retransmissions, jitter | iperf3 -c <IP> -t 10 |
| ping | Latence, perte, MTU | ping -s <size> -c 10 |
| ip | MTU interfaces | ip link show |
| kubectl | Pods, logs | kubectl get pods -w |

## IV. Résultats Expérimentaux

### 4.1 État Initial & Traces Résiduelles

- Interfaces flannel.1 et cni0 présentes malgré nettoyage.
- Nécessité d'un nettoyage agressif pour tests isolés multi-CNI.



```
ubuntu@k8s-worker-1:~$ ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:15:60:4b brd ff:ff:ff:ff:ff:ff
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP mode DEFAULT group default qlen 1000
    link/ether 08:00:27:25:83:2a brd ff:ff:ff:ff:ff:ff
4: flannel.1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue state UNKNOWN mode DEFAULT group default
    link/ether 42:70:76:70:ac:a0 brd ff:ff:ff:ff:ff:ff
39: cni0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue state UP mode DEFAULT group default qlen 1000
    link/ether fa:4d:0f:be:28:c1 brd ff:ff:ff:ff:ff:ff
41: veth3c6260f5@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue master cni0 state UP mode DEFAULT group d
efault
    link/ether 9e:d9:e2:0d:3b:9f brd ff:ff:ff:ff:ff:ff link-netns cni-fd1b6537-4f2a-4efc-8b7f-4cd0db0b4b3e
```

**Figure 1**

### 4.2 Benchmark Applicatif

- Apache Bench (Nginx) baseline : 652 req/sec, 15.3 ms

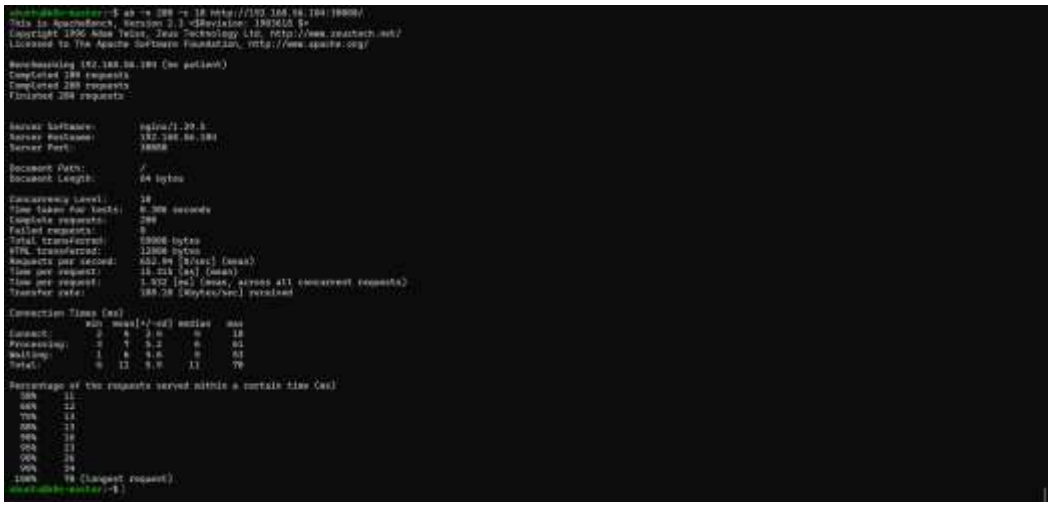


**Figure 2 – Benchmark HTTP**

## 4.3 Cilium – eBPF

- TCP throughput : **2.28 Gbps**, 0 retransmission
- Latence moyenne : ~3 ms, jitter ~1 ms
- Stabilité maximale, bypass d'iptables → routage direct

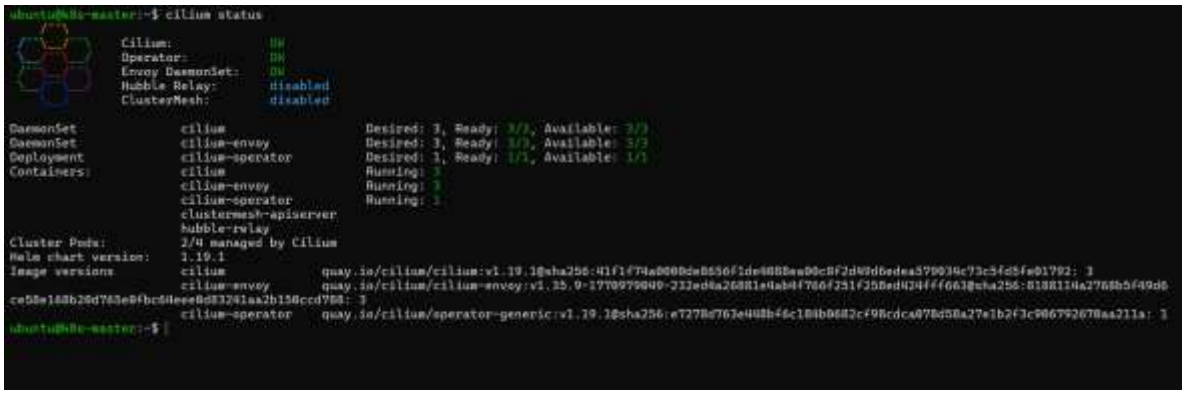


**Figure 3 – Dashboard Cilium**

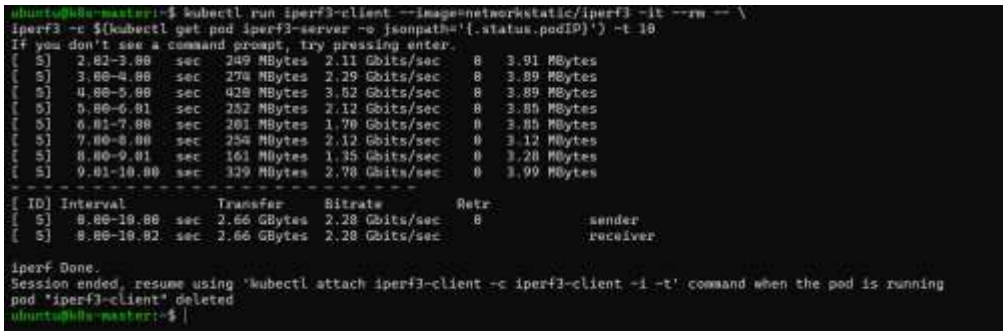


**Figure 4 – Test iperf3 TCP**

## 4.4 Calico – VXLAN/IPRouting

- TCP throughput : 377 Mbps, 169 retransmissions
- Latence moyenne 4.36 ms, jitter 7.45 ms
- MTU 1480 → fragmentation logicielle

```
ubuntu@k8s-master:~$ kubectl get pods -n kube-system -w
NAME                                       READY   STATUS              RESTARTS     AGE
calico-kube-controllers-6879d4fcdc-fdzxc   0/1     ContainerCreating   0            14m
calico-node-5plqc                          1/1     Running             0            79s
calico-node-g4rj8                          1/1     Running             0            7m2s
calico-node-jgjz5                          0/1     Init:1/3            1 (4s ago)   68s
coredns-7c65d6cfc9-29s47                   0/1     Running             0            15m
coredns-7c65d6cfc9-wjtw9                   1/1     Running             0            15m
etcd-k8s-master                            1/1     Running             39           15m
kube-apiserver-k8s-master                  1/1     Running             40           15m
kube-controller-manager-k8s-master         1/1     Running             18           15m
kube-proxy-g4s2h                           1/1     Running             0            68s
kube-proxy-kckh4                           1/1     Running             0            15m
kube-proxy-svwzq                           1/1     Running             0            79s
kube-scheduler-k8s-master                  1/1     Running             37           15m
coredns-7c65d6cfc9-29s47                   1/1     Running             0            15m
calico-node-jgjz5                          0/1     Init:2/3            1 (5s ago)   69s
calico-kube-controllers-6879d4fcdc-fdzxc   0/1     Running             0            14m
calico-kube-controllers-6879d4fcdc-fdzxc   1/1     Running             0            14m
```

**Figure 5 – Pods Calico en Running**

```
ubuntu@k8s-master:~$ kubectl get pods -n kube-system -w
NAME                                       READY   STATUS              RESTARTS     AGE
calico-kube-controllers-6879d4fcdc-fdzxc   0/1     ContainerCreating   0            14m
calico-node-5plqc                          1/1     Running             0            79s
calico-node-g4rj8                          1/1     Running             0            7m2s
calico-node-jgjz5                          0/1     Init:1/3            1 (4s ago)   68s
coredns-7c65d6cfc9-29s47                   0/1     Running             0            15m
coredns-7c65d6cfc9-wjtw9                   1/1     Running             0            15m
etcd-k8s-master                            1/1     Running             39           15m
kube-apiserver-k8s-master                  1/1     Running             40           15m
kube-controller-manager-k8s-master         1/1     Running             18           15m
kube-proxy-g4s2h                           1/1     Running             0            68s
kube-proxy-kckh4                           1/1     Running             0            15m
kube-proxy-svwzq                           1/1     Running             0            79s
kube-scheduler-k8s-master                  1/1     Running             37           15m
coredns-7c65d6cfc9-29s47                   1/1     Running             0            15m
calico-node-jgjz5                          0/1     Init:2/3            1 (5s ago)   69s
calico-kube-controllers-6879d4fcdc-fdzxc   0/1     Running             0            14m
calico-kube-controllers-6879d4fcdc-fdzxc   1/1     Running             0            14m
calico-node-jgjz5                          0/1     Init:2/3            1            92s
calico-node-jgjz5                          0/1     PodInitializing     0            93s
calico-node-jgjz5                          0/1     Running             0            94s
calico-node-jgjz5                          1/1     Running             0            101s
^Cubuntu@k8s-master:~$ kubectl get pods -n kube-system
NAME                                       READY   STATUS    RESTARTS   AGE
calico-kube-controllers-6879d4fcdc-fdzxc   1/1     Running   0          15m
calico-node-5plqc                          1/1     Running   0          2m11s
calico-node-g4rj8                          1/1     Running   0          7m54s
calico-node-jgjz5                          1/1     Running   0          2m
coredns-7c65d6cfc9-29s47                   1/1     Running   0          16m
coredns-7c65d6cfc9-wjtw9                   1/1     Running   0          16m
etcd-k8s-master                            1/1     Running   39         16m
kube-apiserver-k8s-master                  1/1     Running   40         16m
kube-controller-manager-k8s-master         1/1     Running   18         16m
kube-proxy-g4s2h                           1/1     Running   0          2m
kube-proxy-kckh4                           1/1     Running   0          16m
kube-proxy-svwzq                           1/1     Running   0          2m11s
kube-scheduler-k8s-master                  1/1     Running   37         16m
ubuntu@k8s-master:~$
```

**Figure 6 – Stabilisation kube-system**

```
ubuntu@k8s-master:~$ kubectl run iperf3-client --image=networkstatic/iperf3 -it --rm -- \
iperf3 -c $(kubectl get pod iperf3-server -o jsonpath='{.status.podIP}') -t 10
If you don't see a command prompt, try pressing enter.
[  5]   1.02-2.05   sec  38.8 MBytes   318 Mbits/sec    4    798 KBytes
[  5]   2.05-3.01   sec  58.8 MBytes   511 Mbits/sec    0    810 KBytes
[  5]   3.01-4.03   sec  52.5 MBytes   434 Mbits/sec   27    609 KBytes
[  5]   4.03-5.00   sec  43.8 MBytes   377 Mbits/sec   57    639 KBytes
[  5]   5.00-6.00   sec  33.8 MBytes   283 Mbits/sec   25    665 KBytes
[  5]   6.00-7.00   sec  70.0 MBytes   587 Mbits/sec    0    694 KBytes
[  5]   7.00-8.00   sec  30.0 MBytes   252 Mbits/sec   29    710 KBytes
[  5]   8.00-9.00   sec  53.8 MBytes   451 Mbits/sec   26    738 KBytes
[  5]   9.00-10.00  sec  52.5 MBytes   440 Mbits/sec    1    766 KBytes
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Retr
[  5]   0.00-10.00  sec   453 MBytes   380 Mbits/sec  169             sender
[  5]   0.00-10.01  sec   450 MBytes   377 Mbits/sec                  receiver

iperf Done.
Session ended, resume using 'kubectl attach iperf3-client -c iperf3-client -i -t' command when the pod is running
pod "iperf3-client" deleted
ubuntu@k8s-master:~$
```

**Figure 7 – Test iperf3 TCP**

```
ubuntu@k8s-master:~$ kubectl run -i --tty --rm debug --image=busybox --restart=Never -- \
ping -c 10 $(kubectl get pod iperf3-server -o jsonpath='{.status.podIP}')
If you don't see a command prompt, try pressing enter.
64 bytes from 192.168.235.132: seq=2 ttl=62 time=3.085 ms
64 bytes from 192.168.235.132: seq=3 ttl=62 time=3.410 ms
64 bytes from 192.168.235.132: seq=4 ttl=62 time=5.324 ms
64 bytes from 192.168.235.132: seq=5 ttl=62 time=3.164 ms
64 bytes from 192.168.235.132: seq=6 ttl=62 time=3.774 ms
64 bytes from 192.168.235.132: seq=7 ttl=62 time=10.536 ms
64 bytes from 192.168.235.132: seq=8 ttl=62 time=3.456 ms
64 bytes from 192.168.235.132: seq=9 ttl=62 time=3.100 ms

--- 192.168.235.132 ping statistics ---
10 packets transmitted, 10 packets received, 0% packet loss
round-trip min/avg/max = 3.085/4.355/10.536 ms
pod "debug" deleted
ubuntu@k8s-master:~$
```

**Figure 8 – Test ping ICMP**

```
ubuntu@k8s-master:~$ ip addr | grep -A 1 "cali"
3: cali596bcf58a91@if4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1480 qdisc noqueue state UP group default
    link/ether ee:ee:ee:ee:ee:ee brd ff:ff:ff:ff:ff:ff link-netns cni-cd9db764-0237-b8f9-740c-8253a67fc113
--
11: cali0d3f371df11@if4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1480 qdisc noqueue state UP group default
    link/ether ee:ee:ee:ee:ee:ee brd ff:ff:ff:ff:ff:ff link-netns cni-adb0bbf2-1261-0972-5372-08e5ade7ad68
--
14: calid7a5ccee123@if4: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1480 qdisc noqueue state UP group default
    link/ether ee:ee:ee:ee:ee:ee brd ff:ff:ff:ff:ff:ff link-netns cni-f98589ef-91ae-b055-573e-81e938834cf1
ubuntu@k8s-master:~$
```

**Figure 9 – Interfaces veth Calico**

## 4.5 Kube-OVN – Pollution du Noyau

- Pods instables, redémarrages fréquents
- UDP throughput : 2.74–3.5 Mbps, perte 32 % en charge, jitter 4.9 ms
- Latence max ping 1400 octets : 98 ms

```
ubuntu@k8s-master:~$ kubectl get pods -A -w
NAMESPACE     NAME                                       READY   STATUS    RESTARTS        AGE
kube-system   etcd-k8s-master                            1/1     Running   39              163m
kube-system   kube-apiserver-k8s-master                  1/1     Running   40              163m
kube-system   kube-controller-manager-k8s-master         0/1     Running   20 (37s ago)    163m
kube-system   kube-ovn-cni-56d6s                         1/1     Running   1 (3m22s ago)   5m31s
kube-system   kube-ovn-cni-kdh67                         1/1     Running   0               5m29s
kube-system   kube-ovn-cni-rrbbk                         1/1     Running   0               5m37s
kube-system   kube-ovn-controller-6c8d6b5649-nvj5t       1/1     Running   0               8m29s
kube-system   kube-ovn-monitor-7cf56c9b7-g2pqk           1/1     Running   0               19m
kube-system   kube-proxy-7mdfh                           1/1     Running   0               9m59s
kube-system   kube-proxy-cgm69                           1/1     Running   0               9m51s
kube-system   kube-proxy-np784                           1/1     Running   0               9m55s
kube-system   kube-scheduler-k8s-master                  1/1     Running   39 (5m1s ago)   163m
kube-system   ovn-central-854f8bb758-rrw8p               1/1     Running   0               30m
kube-system   ovs-ovn-bqq7h                              1/1     Running   0               36m
kube-system   ovs-ovn-smsnz                              1/1     Running   0               36m
kube-system   ovs-ovn-ts5j2                              1/1     Running   0               36m
```

**Figure 10 – État des pods Kube-OVN**

```
ubuntu@k8s-master:~$ kubectl run iperf3-client --image=networkstatic/iperf3 -it --rm --overrides='{"spec": {"nodeSelector": {"kubernetes.io/host
name": "worker2"}}}' -- iperf3 -c 10.16.0.4 -u -b 100M -t 10
If you don't see a command prompt, try pressing enter.
[  5]   3.02-4.00   sec   191 KBytes  1.59 Mbits/sec  145
[  5]   4.00-5.00   sec   220 KBytes  1.80 Mbits/sec  167
[  5]   5.00-6.00   sec   792 KBytes  6.58 Mbits/sec  602
[  5]   6.00-7.00   sec   275 KBytes  2.25 Mbits/sec  209
[  5]   7.00-8.00   sec   309 KBytes  2.54 Mbits/sec  235
[  5]   8.00-9.01   sec   136 KBytes  1.10 Mbits/sec  103
[  5]   9.01-10.00  sec   136 KBytes  1.12 Mbits/sec  103
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  3.32 MBytes  2.78 Mbits/sec  0.800 ms  0/2502 (0%)  sender
[  5]   0.00-10.07  sec  3.28 MBytes  2.74 Mbits/sec  4.939 ms  26/2500 (1%)  receiver

iperf Done.
Session ended, resume using 'kubectl attach iperf3-client -c iperf3-client -i -t' command when the pod is running
pod "iperf3-client" deleted
```

**Figure 11 – Test UDP iperf3**

```
ubuntu@k8s-master:~$ kubectl run ping-test --image=alpine -it --rm --overrides='{"spec": {"nodeSelector": {"kubernetes.io/hostname": "worker2"}}
}' -- ping -s 1400 -c 5 10.16.0.4
If you don't see a command prompt, try pressing enter.
warning: couldn't attach to pod/ping-test, falling back to streaming logs: Internal error occurred: Internal error occurred: error attaching to
container: failed to load task: no running task found: task 7e1ea55a3e98b2290320fed8f7a4ebfe5922b2ef8cc73dd5ef62c4c77efb939a not found: not foun
d
PING 10.16.0.4 (10.16.0.4): 1400 data bytes
1408 bytes from 10.16.0.4: seq=0 ttl=64 time=98.627 ms
1408 bytes from 10.16.0.4: seq=1 ttl=64 time=36.866 ms
1408 bytes from 10.16.0.4: seq=2 ttl=64 time=20.243 ms
1408 bytes from 10.16.0.4: seq=3 ttl=64 time=14.480 ms
1408 bytes from 10.16.0.4: seq=4 ttl=64 time=10.162 ms

--- 10.16.0.4 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 10.162/36.875/98.627 ms
pod "ping-test" deleted
```

**Figure 12 – Ping MTU 1400 octets**

## 4.6 Flannel – Crashes et Limites

- Pods Flannel en CrashLoopBackOff
- MTU 1450 → fragmentation, instabilité

```
ubuntu@k8s-master:~$ kubectl get pods -n kube-flannel
NAME                    READY   STATUS             RESTARTS      AGE
kube-flannel-ds-6q8zf   0/1     CrashLoopBackOff   4 (39s ago)   3m8s
kube-flannel-ds-bcplx   0/1     CrashLoopBackOff   4 (35s ago)   3m8s
kube-flannel-ds-mm5f2   0/1     CrashLoopBackOff   3 (45s ago)   3m8s
ubuntu@k8s-master:~$
```

**Figure 13 – CrashLoopBackOff Flannel**

### 4.7 Synthèse Quantitative

| CNI | TCP Débit | Retransmissions | Latence Moy. | Jitter | Perte UDP | MTU |
|---|---|---|---|---|---|---|
| Cilium | 2.28 Gbps | 0 | 3.0 ms | ~1 ms | N/A | 1500 |
| Calico | 377 Mbps | 169 | 4.36 ms | 7.45 ms | N/A | 1480 |
| Kube-OVN | 3.5 Mbps | Élevées | 36 ms | 88.5 ms | 32 % | 1400 |
| Flannel | Échec | - | - | - | - | 1450 |

## V. Discussion

- **Cilium :** supérieur, faible latence, stable, eBPF bypass efficace
- **Calico :** stable mais overhead VXLAN/iptables limite débit
- **Kube-OVN :** multi-CNI → pollution noyau, perte UDP, latence extrême
- **Flannel :** instable, MTU réduit, inadéquat en production

**Implications pratiques :**

- Nettoyage strict des clusters pour benchmarks
- Planification MTU adaptée au CNI
- Monitoring retransmissions TCP et jitter

**Limitations :**

- Cluster 3 nœuds, charge synthétique
- Absence de NetworkPolicy
- Homogénéité matérielle non testée

# VI. Conclusion

| Besoin | CNI Recommandé | Justification |
|---|---|---|
| Haute performance | Cilium | 2.28 Gbps, 0 retransmission |
| Stabilité opérationnelle | Calico | Maturité, support entreprise |
| À éviter en production | Flannel | CrashLoopBackOff, MTU réduit |
| Surveillance avancée nécessaire | Kube-OVN | Pollution noyau, perte UDP |

**Contributions scientifiques :**

1. Quantification de la pollution du noyau par CNIs multiples
2. Méthodologie reproductible avec nettoyage forensique
3. Base de données comparative pour 4 CNI majeurs

# VII. Travaux Futurs

1. Benchmark sur 50+ nœuds
2. Impact des NetworkPolicies
3. Analyse forensique eBPF des chemins de paquets
4. Benchmarks applicatifs réels : Redis, Kafka, MySQL

# VIII. Références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2. Calico. (2025). Project Calico Documentation
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5. iPerf3. (2025). iPerf3 Documentation